Smart Healthcare Monitoring and Medication Management System

Project Report

Submitted by

Ardra P

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1 Abstract

This project presents a Smart Healthcare Monitoring and Medication Management System implemented using the ESP32 microcontroller, multiple sensors, FreeRTOS, MQTT communication, and Adafruit IO dashboards. The system continuously monitors patient vital signs and environmental parameters, generates medical and facility alerts, performs periodic data aggregation, and enables remote medication dosage adjustment with safety mechanisms. The objective is to simulate an intelligent hospital environment that improves patient safety, real-time monitoring, and remote control capabilities.

2 Objectives

The objectives of this project are:

- Monitor patient vital signs in real time.
- Detect abnormal medical conditions and generate alerts.
- Monitor environmental conditions inside healthcare facilities.
- Aggregate sensor readings for better analysis.
- Enable remote medication dosage control.
- Implement safety mechanisms using LEDs and buzzers.

3 Task 1: Smart Patient and Facility Monitoring

3.1 Procedure

- Initialize the ESP32, required sensors, OLED display, MQTT client, queues, semaphores, mutexes, and watchdog timer.
- Create separate FreeRTOS tasks for patient vital monitoring, environmental monitoring, MQTT communication, alert handling, and display management using priority-based scheduling.
- Implement additional health monitoring tasks for parameters such as blood pressure and ECG, ensuring that each parameter is processed independently.
- Collect sensor readings periodically and transfer the data between tasks using FreeRTOS queues.
- Use semaphores to synchronize task execution and mutexes to protect shared resources, such as the OLED display and communication interfaces, thereby preventing race conditions.
- Process the acquired data to detect abnormal medical and environmental conditions and generate appropriate alerts when predefined thresholds are exceeded.
- Publish sensor data and alert information to the cloud dashboard through MQTT communication for remote monitoring.

- Update the OLED display with real-time patient information, environmental conditions, and system status.
- Continuously reset and monitor the watchdog timer to ensure system reliability and recover from potential task failures or deadlocks.
- Verify that all tasks execute concurrently and maintain real-time responsiveness, ensuring stable and reliable healthcare monitoring operation.

3.2 observation

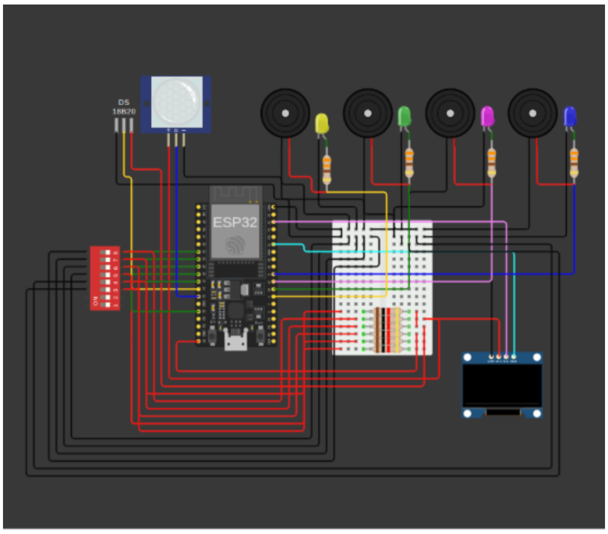


Figure 1: circuit image

Figure 2: dashboard

- Patient vital signs, including temperature, heart rate, SpO_2, blood pressure, and ECG, were successfully monitored in real time.
- Medical alerts were generated automatically whenever the measured values exceeded predefined threshold limits.
- Environmental parameters such as room temperature, oxygen level, air quality index (AQI), and motion detection were continuously monitored.

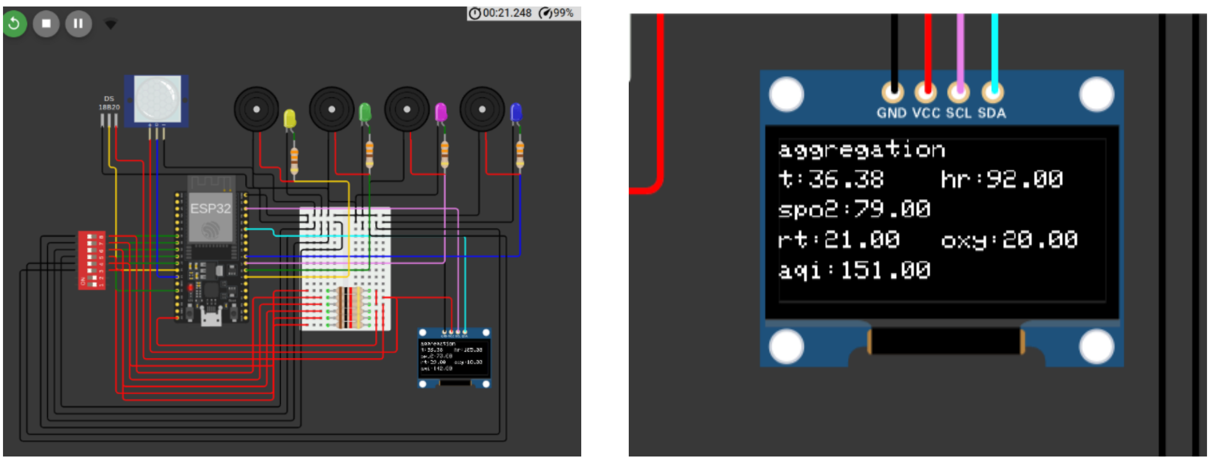
- Facility alerts were triggered under abnormal environmental conditions, ensuring improved safety and awareness.
- FreeRTOS tasks executed concurrently without affecting overall system responsiveness and performance.
- Queues, semaphores, and mutexes provided reliable inter-task communication and prevented race conditions.
- Sensor data and alert information were successfully transmitted to the Adafruit IO dashboard using MQTT communication.
- The watchdog timer enhanced system reliability by detecting and recovering from unresponsive task behavior.

4 Task 2: Dual-Role IoT Monitoring System with Role-Based Dashboards

4.1 Procedure

- Create separate MQTT feeds for patient vitals and environmental parameters.
- Publish heart rate, SpO_2, and body temperature data to the Medical Staff Dashboard.
- Publish room temperature, oxygen level, and AQI data to the Facility Management Dashboard.
- Configure gauge widgets with predefined safe operating thresholds.
- Generate medical and facility alerts whenever abnormal conditions are detected.
- Aggregate sensor readings periodically to provide summarized reports.

4.2 Observation



(a) Circuit Diagram

(b) OLED Display

Figure 3: Task 2 Implementation Results

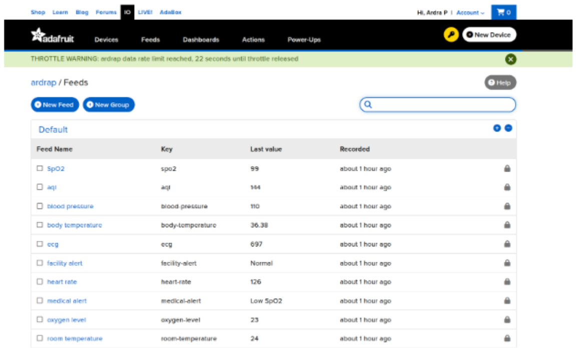


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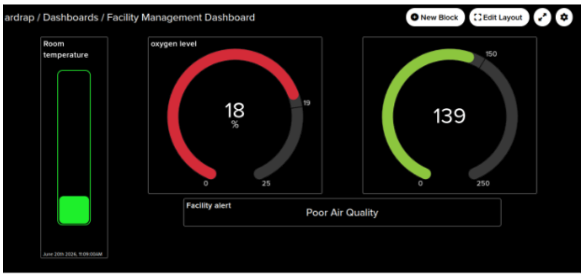
Figure 4: feeds



(a) Facility Management Dashboard

(b) Medical Staff Dashboard

Figure 5: Role-Based IoT Dashboards

- Patient and environmental data were successfully separated into dedicated dashboards.
- Real-time monitoring and alert generation improved system visibility and safety.
- Periodic aggregation provided concise summaries of sensor data.
- The role-based architecture enhanced scalability and resembled enterprise IoT deployments.

5 Conclusion

The project successfully integrates patient monitoring, environmental sensing, data aggregation, and medication dosage management into a unified IoT-based healthcare platform.

The combination of ESP32, MQTT, Adafruit IO, and FreeRTOS enables efficient real-time operation while ensuring patient safety through multiple alert mechanisms and controlled medication delivery.

The system demonstrates the potential of intelligent healthcare technologies in supporting remote monitoring and smart decision-making.